

RTS,S IMPLEMENTATION COVERAGE AND ILLNESS-AVERTED ESTIMATES ACROSS FIVE SUB-SAHARAN AFRICAN SITES

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BACKGROUND

RTS,S/AS01 is the first WHO-recommended pediatric malaria vaccine, with approximately 36 percent efficacy against clinical malaria in Phase 3. Real-world implementation outcomes depend on coverage. We report multi-site coverage and coverage-adjusted illness-averted estimates from a prospective observational implementation cohort across five sub-Saharan African sites, with two-source verification applied at each site.

METHODS

- Cohort: children in eligible birth-cohort window under Amendment 4 of the multi-site implementation protocol.
- Coverage: doses administered divided by eligible birth cohort.
- Two-source verification: raw field-data pipeline (Dropbox) reconciled against site-coordinator partner sign-off (Salesforce).
- Winner rule when sources disagree: newest and closest to source.
- Illness averted: sum across sites of coverage x birth cohort x efficacy fraction (0.36) x baseline malaria incidence (312 per 1000, site-averaged).
- Sites where sources materially disagree are held as open conclusions rather than forced to verdict.

RESULTS

Per-site coverage
Kintampo Ghana: 68.4% N = 892 eligible cohort | 611 doses administered | **LOCKED**
Kilifi Kenya: 79.6% N = 1,024 eligible cohort | 815 doses administered | **OPEN CONCLUSION**
(pipeline encounter count exceeds partner sign-off by 33 doses)
Manhica Mozambique: 81.1% N = 745 eligible cohort | 604 doses administered | **LOCKED**
Nanoro Burkina Faso: 73.4% N = 981 eligible cohort | 720 doses administered | **LOCKED**
Bagamoyo Tanzania: 85.4% N = 740 eligible cohort | 632 doses administered | **LOCKED**
Aggregate weighted coverage: 77.2 percent across 4,382 enrolled encounters.

COVERAGE-ADJUSTED ILLNESS AVERTED

Aggregate point estimate: 380 clinical malaria cases averted across the cohort at current coverage.
Range accounting for Kilifi open conclusion: 361 to 401.
Per-site contribution:
Kintampo Ghana: 68.5 cases averted
Kilifi Kenya: 91.6 cases averted (range 87.8 to 95.4)
Manhica Mozambique: 67.9 cases averted
Nanoro Burkina Faso: 80.9 cases averted
Bagamoyo Tanzania: 70.9 cases averted

LIMITATIONS

One site holds an open conclusion due to source disagreement (Kilifi). We present as range rather than force a verdict. Baseline malaria incidence held site-averaged rather than site-specific in this pass. Efficacy fraction held constant across sites at 0.36; site-specific efficacy is not currently estimable from this cohort. Cross-source reconciliation surfaced stale reference documents (Confluence protocol v3.2, Airtable manuscript pipeline row) that we set aside.

IMPLICATIONS

Four of five implementation sites clear coverage thresholds associated with meaningful population-level illness averted. Multi-source reconciliation is not project-management overhead - it is the analysis. The methodology transfers to other pediatric vaccines with delivery-heterogeneity.

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